

Machine Learning Project Report: Bank Marketing Campaign Prediction

Author: Roshan Shakya

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Executive Summary

This project analyzes customer subscription patterns to term deposits through the Bank Marketing Dataset. The research aims to enhance marketing campaign success through improved customer identification methods, which utilize data analysis techniques.

The dataset contains customer demographic details together with their financial information and data about campaign activities. The data preprocessing process required Rapid Miner to perform three tasks, which included designating roles, converting nominal data to numerical format and normalizing data for modeling purposes.

The multiple machine learning systems, which included Logistic Regression, Linear Discriminant Analysis (LDA), K-Nearest Neighbors, Naive Bayes, Bagging (Random Forest) and Boosting ensemble techniques were applied. Use of accuracy, confusion matrix and ROC-AUC metrics together with cross-validation to evaluate model performance.

The results show that ensemble methods achieved superior performance through Random Forest and Boosting because these methods can process complex patterns while decreasing overfitting problems. Exploratory Data Analysis showed that balance and job type variables affect customer behavior while the analysis identified major outliers and distribution patterns, which were not normal.

The project demonstrates that machine-learning methods enable financial institutions to create focused marketing campaigns, which result in improved customer acquisition and decreased campaign expenses and better decision-making processes.

Introduction

Currently, banks encounter more difficulties to promote their products throughout the customer base because of the present day competitive financial marketplace. Banks provide term deposits as their primary product, which requires customers to keep their money invested during a predetermined time. However, identifying potential customers who are likely to subscribe remains a critical challenge.

This project focuses on predicting customer subscription to term deposits using machine-learning techniques. The Bank Marketing Dataset serves as the study dataset, which contains customer demographic data together with their financial information and previous campaign interactions. The target variable is whether a customer subscribes to a term deposit (deposit: yes/no).

The main problem addressed in this project is the inefficiency of traditional marketing campaigns which banks contact a large number of customers without knowing who is most likely to respond positively to their marketing efforts. The process results in higher expenses and decreased effectiveness for marketing campaigns. The project uses machine learning models, which include Logistic Regression and K-Nearest Neighbors (KNN), and Naive Bayes and ensemble techniques, which contain Bagging and Boosting to increase prediction accuracy while enabling data-driven decision-making.

The study demonstrates its value through establishing better marketing techniques, which financial institutions can use to improve their operations. Banks can achieve better customer targeting through accurate prediction models, which will also reduce their operational expenses and enhance their success rate with potential customers. The project shows how Rapid Miner machine-learning tools enable users to process real-world data, which leads to business insights.

Data Description

Dataset Overview

The Bank Marketing Dataset serves as the project dataset, which contains customer data from a financial institution. The dataset serves its main purpose to forecast customer subscription rates for term deposits.

The dataset contains categorical and numerical data, which shows customer demographics and financial details and their responses to marketing campaigns.

Key Features Include:

- Demographic variables: age, job, marital status, education
- Financial variables: balance, housing, loan, deposit
- Campaign-related variables: contact duration, campaign, previous
- Target variable (label): deposit (yes/no) which indicates whether the customer subscribed.

Characteristics of dataset

The dataset contains a mix of nominal and numerical attributes, which makes it appropriate for conducting classification assignments. The binary target variable enables the application of supervised machine learning algorithms, which include Logistic Regression and KNN and Naïve Bayes and ensemble methods.

Data Preprocessing

1. Handling missing values

Missing values were treated through suitable methods, which included substituting missing numerical data with mean values and missing categorical data with mode values to achieve complete data set.

2. Nominal to Numerical conversion

The Nominal to Numerical operator was used to transform categorical variables into numerical data, which machine-learning algorithms could process.

3. Role Assignment

The target variable deposit was assigned as the label using the Set Role operator.

4. Normalization

The Normalize operator converted numerical data into normalized form, which produced standardized value ranges. The procedure enhances KNN and other distance-based algorithms through its implementation.

5. Feature Selection/Cleaning

Attributes were assessed to determine their importance for prediction purposes and decided which attributes to keep.

6. Data Splitting/Validation Setup

The dataset underwent Cross Validation testing which confirmed model performance accuracy while preventing overfitting problems.

Hence, the preprocessing procedures established a clean dataset, which met proper structural requirements for developing accurate machine learning models.

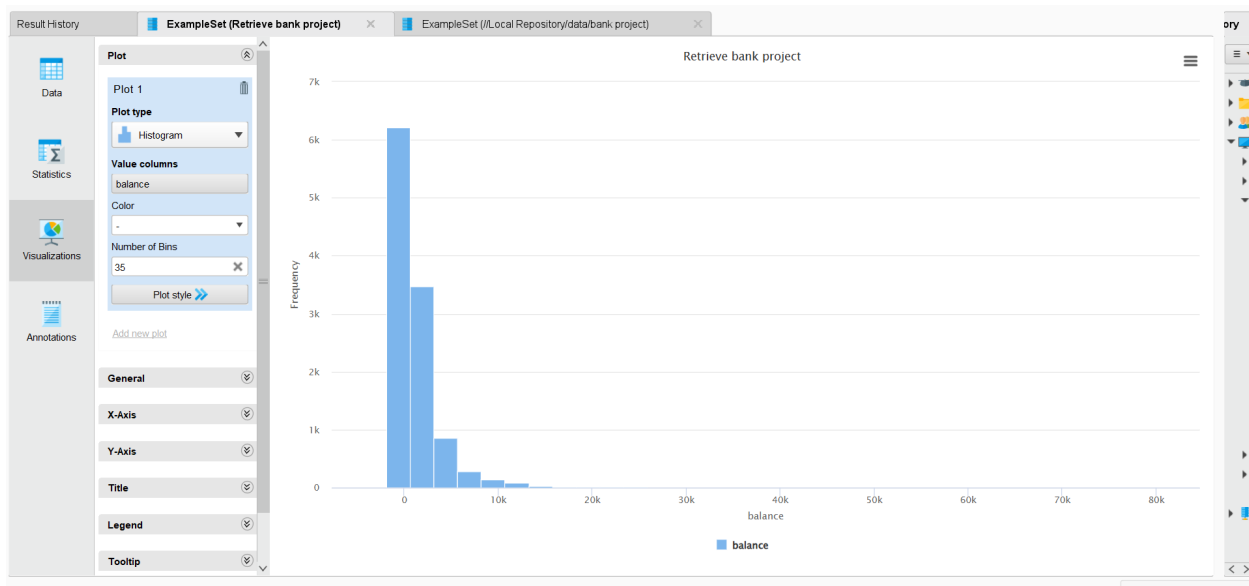
Exploratory Data Analysis

Univariate Analysis

A histogram was created to examine the distribution of the balance variable using 35 bins.

Figure 1

Histogram of Balance



The distribution of balance is highly right-skewed, with most customers having balances close to zero, including some negative values. A small number of customers have very high balances, resulting in a long right tail. This indicates the presence of extreme values (outliers) and a non-normal distribution.

The results show that the majority of customers maintain low account balances, while only a few customers hold significantly higher balances. This uneven distribution suggests variability in customer financial status. Since the project aims to predict deposit subscription, understanding

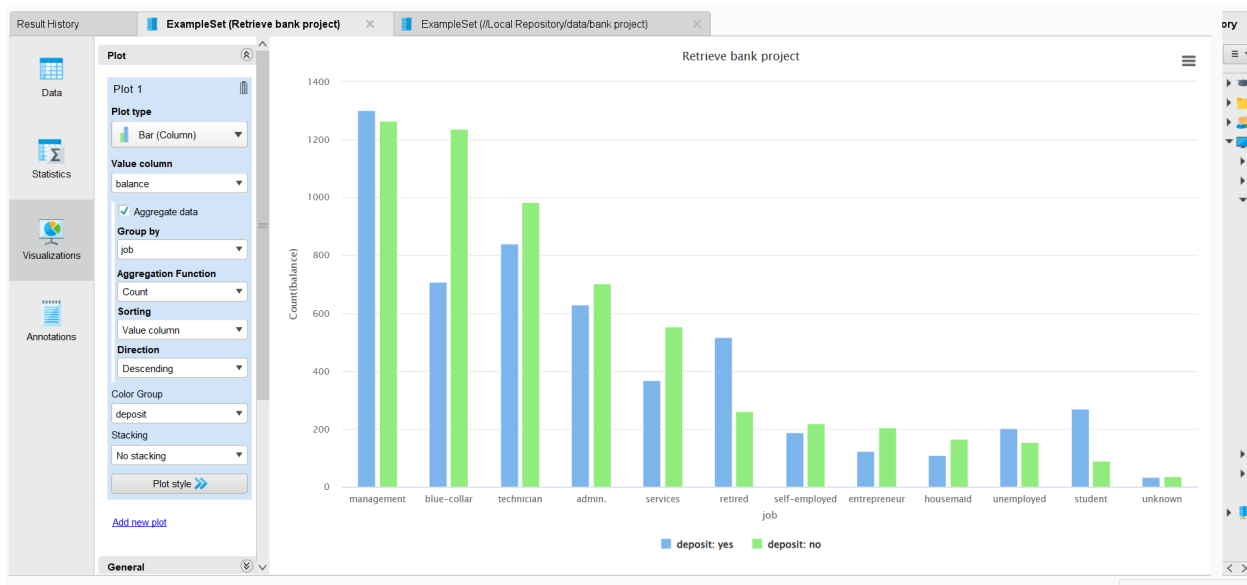
the distribution of balance is important, as it may influence customer financial behavior and decision-making.

Bivariate Analysis

A bar chart was created to analyze the relationship between job category and deposit subscription. The number of customers in each job category was calculated and grouped by deposit outcome (yes/no), and the results were sorted in descending order for better comparison.

Figure 2

Job category vs Deposit Subscription (Bivariate Analysis)



Management has the highest number of customers, with slightly more “yes” than “no”. Blue-collar category shows a high number of customers but is dominated by “no”. Technician and admin categories have relatively high counts of both “yes” and “no”. Retired customers show a noticeably higher proportion of “yes” compared to “no”. Services category has more “no” than

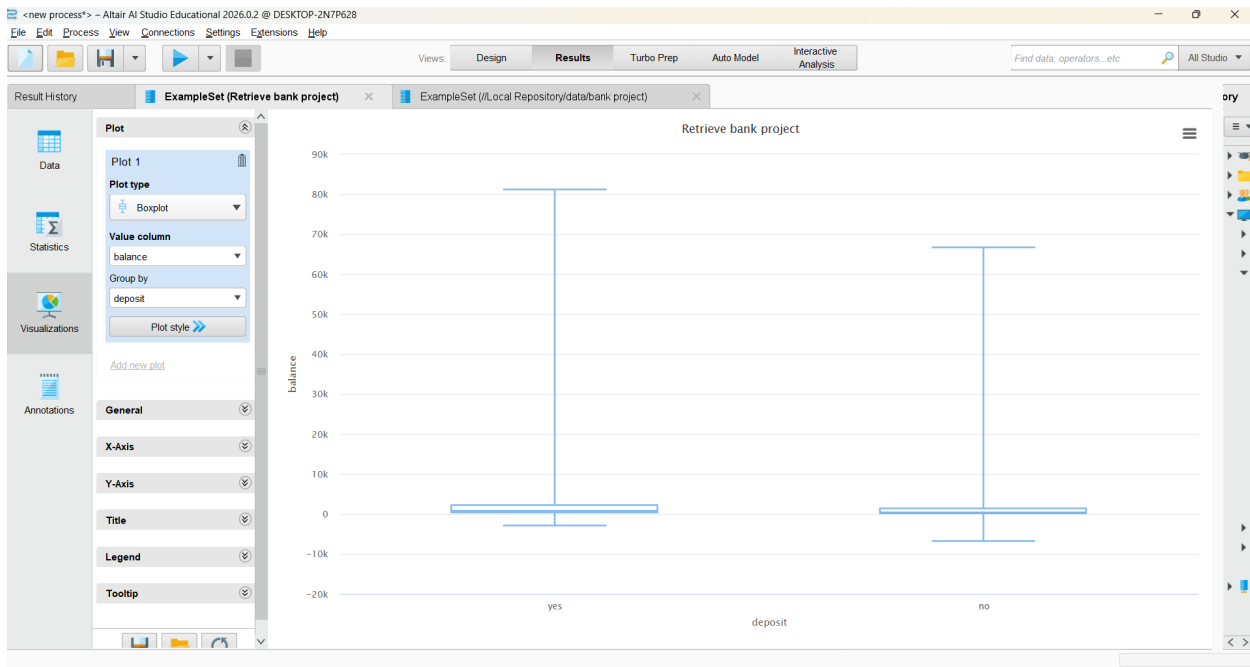
“yes”, indicating lower subscription tendency. Categories such as student, unemployed, and unknown have lower counts overall.

The results indicate that job category influences customer subscription behavior, although the effect varies across different occupations. While most job groups have more non-subscribers, certain categories such as management and retired customers show a relatively higher likelihood of subscribing. This suggests that occupational characteristics may affect financial decisions and investment behavior.

Since the objective of the project is to predict whether a customer will subscribe to a term deposit, the job variable provides meaningful segmentation. It helps identify customer groups with different subscription tendencies, thereby improving the predictive power of classification models.

Outlier Detection

A box plot was used to examine how balance distribution varies between two deposit groups, which include customers who have made deposits, and those who have not. The graph displays information about how data points spread across different distance ranges and calculate their average value while showing which data points exist outside typical patterns.

Figure 3*Box plot balance by deposit (Outlier Detection)*

The box plot shows that the balance variable is highly skewed because most values cluster around zero while only a few values reach maximum heights of 80,000. Both categories (deposit = yes and deposit = no) exhibit long upper whiskers and extreme outliers which indicate the existence of customers who maintain exceptionally high account balances.

The account balance analysis shows that both groups maintain low balance levels, which remain consistent because most customers establish average account balance amounts that stay constant across different subscription types. The presence of outliers in both categories suggests that high-balance customers exist in both subscribers and non-subscribers meaning balance alone may not create strong customer behavior differences.

The balance variable contains significant outliers and a wide range of values, which may affect model performance because algorithms with extreme value sensitivity will struggle to process this data. The researchers should treat these outliers as valuable customer data because they represent important customer segments, which require special observation methods instead of normal removal procedures.

Modeling

Logistic Regression

Overview

Logistic Regression was used as a baseline classification model to predict whether a customer will subscribe to a term deposit (deposit = yes/no). It is widely used for binary classification problems due to its simplicity, interpretability, and efficiency.

Methodology

The model was implemented using 10-fold cross-validation with stratified sampling to ensure reliable and unbiased evaluation. The dataset was first preprocessed by assigning the target variable (deposit) as the label and converting categorical variables using the Nominal to Numerical operator.

Within the cross-validation framework:

- Logistic Regression was applied in the training phase
- The trained model was tested using the Apply Model operator
- Performance was evaluated using the Performance (Binominal Classification) operator

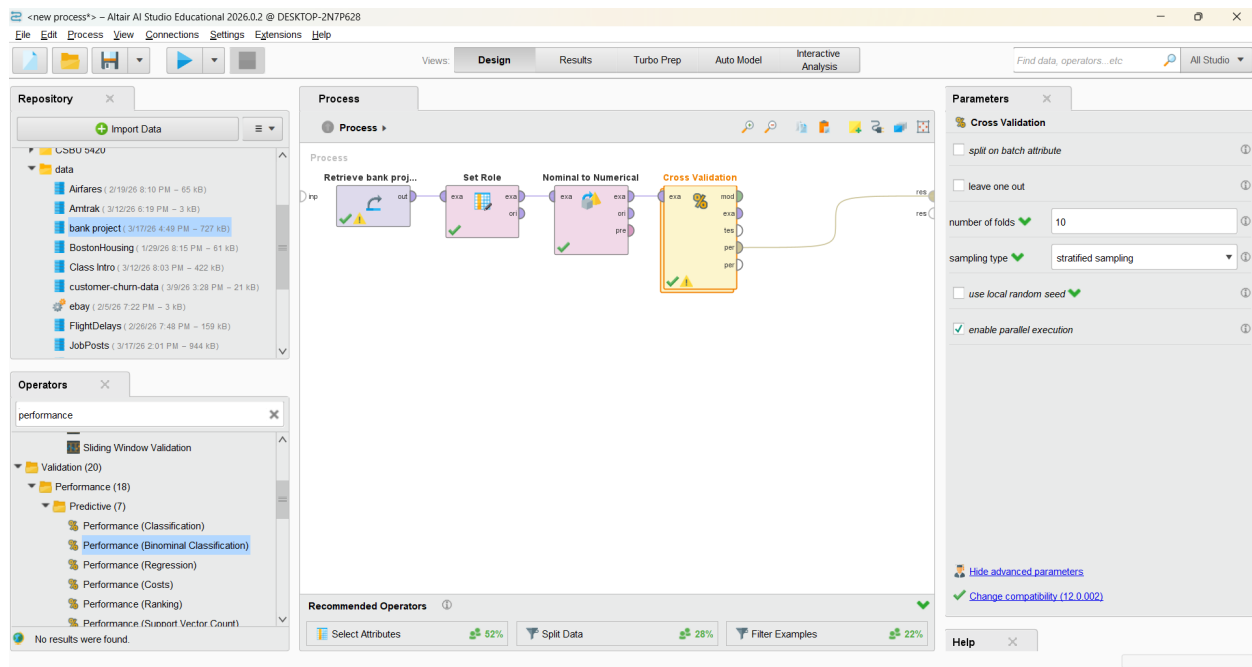
Hyper parameters

- Default Logistic Regression settings
- Cross-validation: 10 folds
- Sampling type: Stratified sampling

- Parallel execution enabled

Figure 4

Rapid Miner process for Logistic Regression



<new process> - Altair AI Studio Educational 2026.0.2 @ DESKTOP-2N7P628

File Edit Process View Connections Settings Extensions Help

Views: Design Results Turbo Prep Auto Model Interactive Analysis

Find data, operators, etc. All Studio

Repository

Import Data

data

- Airfares (2/19/26 8:10 PM - 65 KB)
- Amtrak (3/12/26 6:19 PM - 3 KB)
- bank project (3/17/26 4:49 PM - 727 KB)
- BostonHousing (1/29/26 8:15 PM - 61 KB)
- Class Intro (3/12/26 8:03 PM - 422 KB)
- customer-churn-data (3/9/26 3:28 PM - 21 KB)
- ebay (2/5/26 7:22 PM - 3 KB)
- FlightDelays (2/26/26 7:48 PM - 159 KB)
- JobPosts (3/17/26 2:01 PM - 944 KB)

Operators

performance

- Sliding Window Validation
- Validation (20)
 - Performance (18)
 - Predictive (7)
 - Performance (Classification)
 - Performance (Binominal Classification)
 - Performance (Regression)
 - Performance (Costs)
 - Performance (Ranking)
 - Performance (Support Vector Count)

No results were found.

Process

Process > Cross Validation

Training

Logistic Regression

Testing

Apply Model

Performance

Parameters

Cross Validation

- ☐ split on batch attribute
- ☐ leave one out
- number of folds 10
- sampling type stratified sampling
- ☐ use local random seed
- ☒ enable parallel execution

Hide advanced parameters

Change compatibility (12.0.002)

Help

Recommended Operators

- Select Attributes 52%
- Split Data 28%
- Filter Examples 22%

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Operators

performance

- Sliding Window Validation
- Validation (20)
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 - Predictive (7)
 - Performance (Classification)
 - Performance (Binominal Classification)
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 - Performance (Costs)
 - Performance (Ranking)
 - Performance (Support Vector Count)

No results were found.

Process

Process > Cross Validation

Training

Logistic Regression

Testing

Apply Model

Performance

Parameters

Performance (Performance (Binominal Classification))

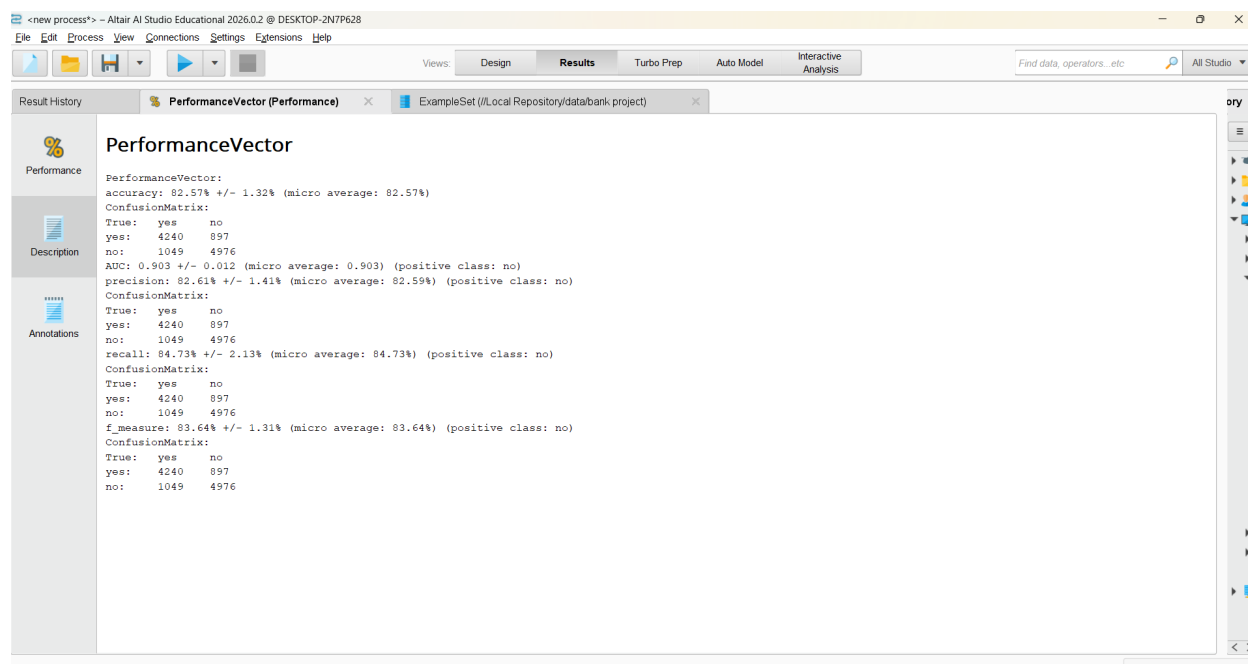
- ☐ manually set positive class
- main criterion accuracy
- ☒ accuracy
- ☐ classification error
- ☐ kappa
- ☐ AUC (optimistic)
- ☒ AUC
- ☐ AUC (pessimistic)
- ☒ precision
- ☒ recall
- ☐ lift
- ☐ fallout
- ☒ f measure

Hide advanced parameters

Help

Recommended Operators

- Select Attributes 52%
- Split Data 28%
- Filter Examples 22%



Performance Result

The Logistic Regression model achieved the following performance:

- Accuracy: 82.57%
- AUC: 0.903
- Precision: 82.61%
- Recall: 84.73%
- F1-score: 83.64%

Confusion matrix

The confusion matrix shows that the model correctly classified a large number of both subscribed and non-subscribed customers. However, there are some misclassifications, indicating that the model struggles with certain complex patterns in the data.

Logistic Regression provides stable and interpretable results, making it useful for understanding the influence of predictors on customer subscription behavior. However, due to its assumption of linear relationships, it is less effective in capturing complex interactions compared to ensemble models. Overall, Logistic Regression delivered moderate to good performance, serving as a strong baseline model.

Limitation of model

- Assumes linear relationships between variables
- Limited ability to capture complex patterns
- Affected by class imbalance (more “no” cases)
- Produces higher false negatives (misses some subscribers)

Linear Discriminant Analysis (LDA)

Overview

Linear Discriminant Analysis (LDA) was used as a classification model to predict whether a customer will subscribe to a term deposit (deposit = yes/no). A statistical method focuses on maximizing the separation between classes.

Methodology

The model was implemented using 10-fold cross-validation with stratified sampling to ensure reliable evaluation. Data preprocessing included assigning the label (deposit), converting categorical variables, and normalizing numerical attributes.

Additionally, Information Gain and Select by Weights (top 15 features) performed feature selection using Weight before applying LDA.

Within the cross-validation:

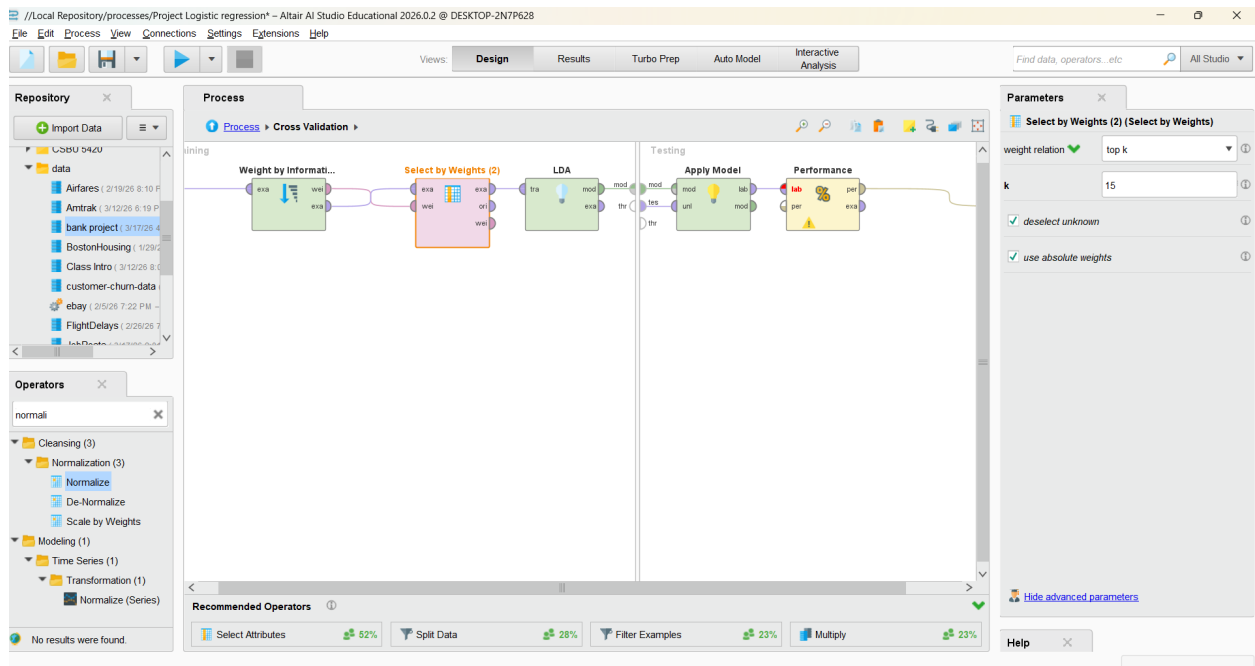
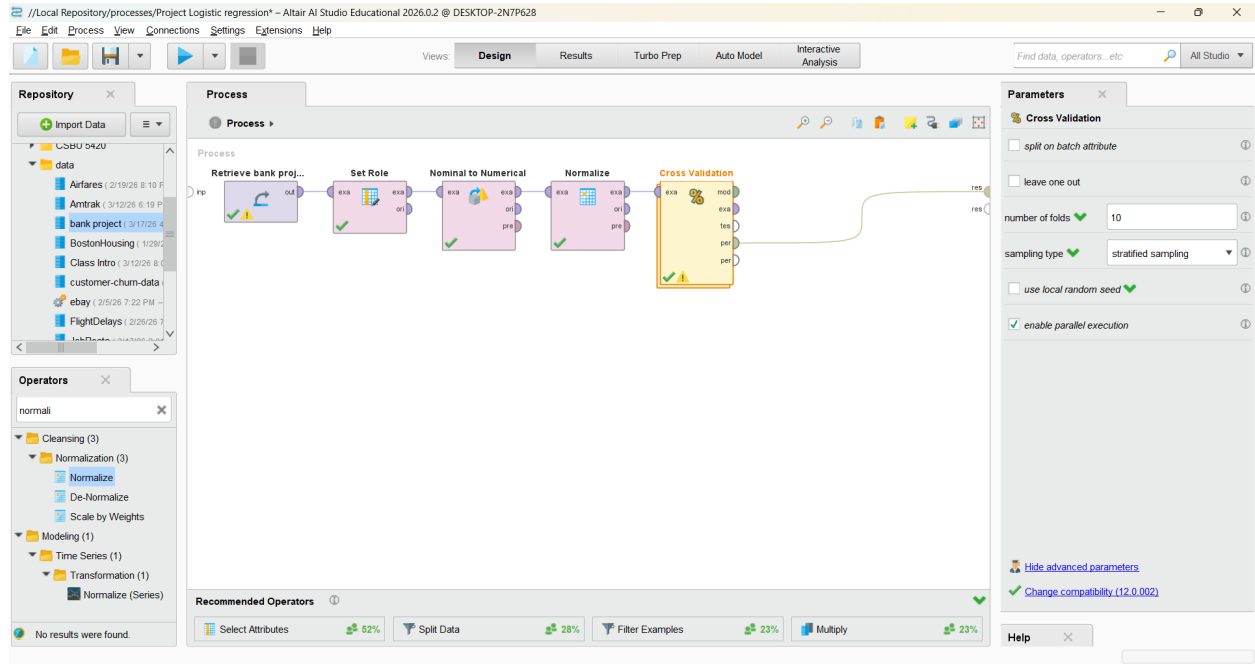
- LDA was trained on selected features
- Apply Model was used for testing
- Performance (Binominal Classification) was used for evaluation

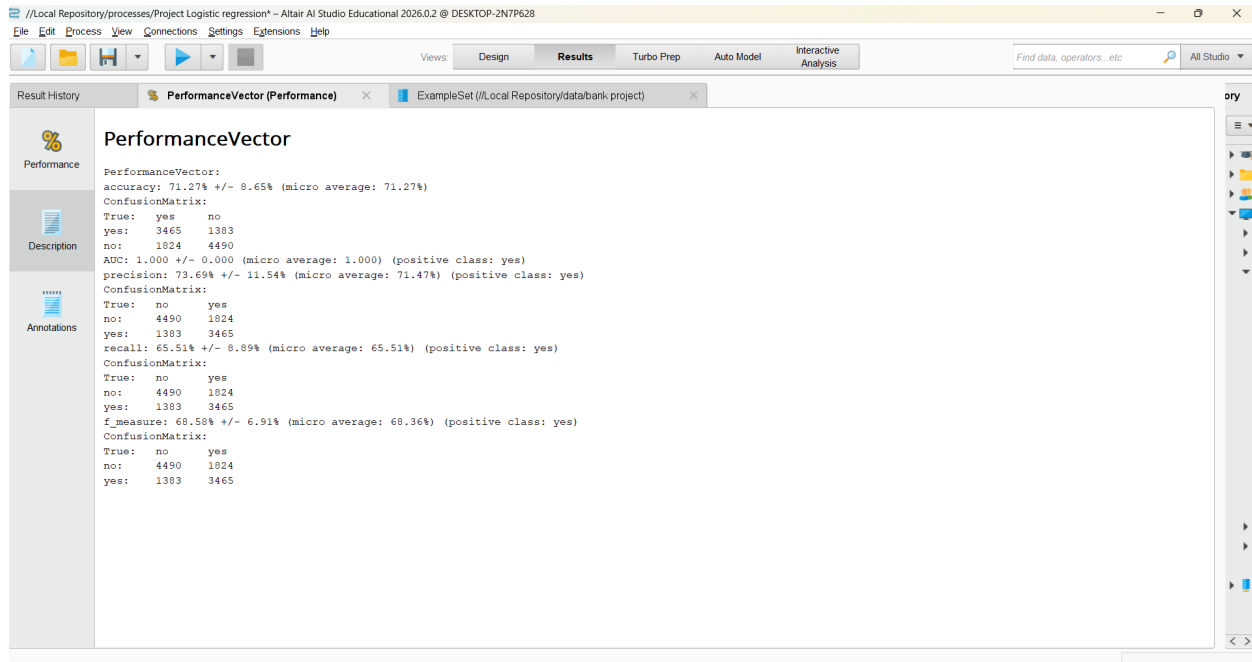
Hyper parameters

- Default LDA settings
- Feature selection: Top 15 attributes
- Cross-validation: 10 folds
- Sampling: Stratified sampling

Figure 5

Rapid miner process for LDA





Performance Result

The LDA model achieved the following performance:

- Accuracy: 71.27%
- AUC: 1.000
- Precision: 73.69%
- Recall: 65.51%
- F1-score: 68.58%

Confusion Matrix

The model correctly classified a reasonable number of customers but showed more misclassifications compared to Logistic Regression. In particular, it failed to identify some actual subscribers, leading to lower recall.

Although LDA attempts to separate classes effectively, its performance is limited by strong assumptions about data distribution. Even with feature selection, it does not capture complex patterns as well as more advanced model. Overall, LDA achieved moderate performance, but lower accuracy compared to Logistic Regression and ensemble models. While it benefits from dimensionality reduction and feature selection, it is less suitable for complex real-world datasets like bank marketing data.

Limitation of Model

- Assumes normal distribution of data
- Assumes equal covariance across classes
- Limited ability to capture complex patterns
- Lower recall (misses some subscribers)

K -NN

Overview

K-Nearest Neighbors (K-NN) was used as a classification model to predict whether a customer will subscribe to a term deposit (deposit = yes/no). A distance-based algorithm classifies a new customer based on the majority class of its nearest neighbors.

Methodology

The model was implemented using 10-fold cross-validation with stratified sampling to ensure reliable and unbiased evaluation.

Data preprocessing steps included:

- Setting the label (deposit)
- Converting categorical variables (Nominal to Numerical)
- Normalizing numerical attributes (important for distance-based models)

Inside cross-validation:

- K-NN model was trained
- Apply Model used for testing
- Performance (Binominal Classification) used for evaluation

Hyper Parameters

- K-NN settings (k=5, distance measure=Euclidean, Measure type=Numerical)
- Distance-based classification
- Cross-validation: 10 folds
- Sampling: Stratified sampling
- Normalization applied (very important for K-NN)

Figure 6

Rapid miner process for K- NN

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Repository

Import Data

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 - data
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 - Amtrak (3/12/26 6:19 PM - 3 KB)
 - bank project (3/17/26 4:49 PM)
 - BostonHousing (1/29/26 8:15)
 - Class Intro (3/12/26 8:03 PM)
 - customer-churn-data (3/9/26)
 - ebay (2/5/26 7:22 PM - 3 KB)
 - FlightDelays (2/26/26 7:48 PM)

Operators

KNN

- Blending (1)
 - Examples (1)
 - Sampling (1)
 - Sample (Kennard-Stone)
- Modeling (1)
 - Predictive (1)
 - Lazy (1)
 - k-NN
- Extensions (1)
 - Text Processing (1)

No results were found.

Process

Process

Retrieve bank proj... Set Role Nominal to Numerical Normalize Cross Validation

Recommended Operators

- Select Attributes 52%
- Split Data 28%
- Filter Examples 23%
- Multiply 23%

Parameters

Process

- logverbosity: init
- logfile:
- resultfile:
- random seed: 2001
- send mail: never
- encoding: SYSTEM

[Hide advanced parameters](#)

[Change compatibility \(12.0.002\)](#)

Help

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Views: Design Results Turbo Prep Auto Model Interactive Analysis

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Repository

Import Data

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 - FlightDelays (2/26/26 7:48 PM)

Operators

KNN

- Blending (1)
 - Examples (1)
 - Sampling (1)
 - Sample (Kennard-Stone)
- Modeling (1)
 - Predictive (1)
 - Lazy (1)
 - k-NN
- Extensions (1)
 - Text Processing (1)

No results were found.

Process

Process > Cross Validation

Training: k-NN

Testing: Apply Model Performance

Recommended Operators

- Select Attributes 52%
- Split Data 28%
- Filter Examples 23%
- Multiply 23%

Parameters

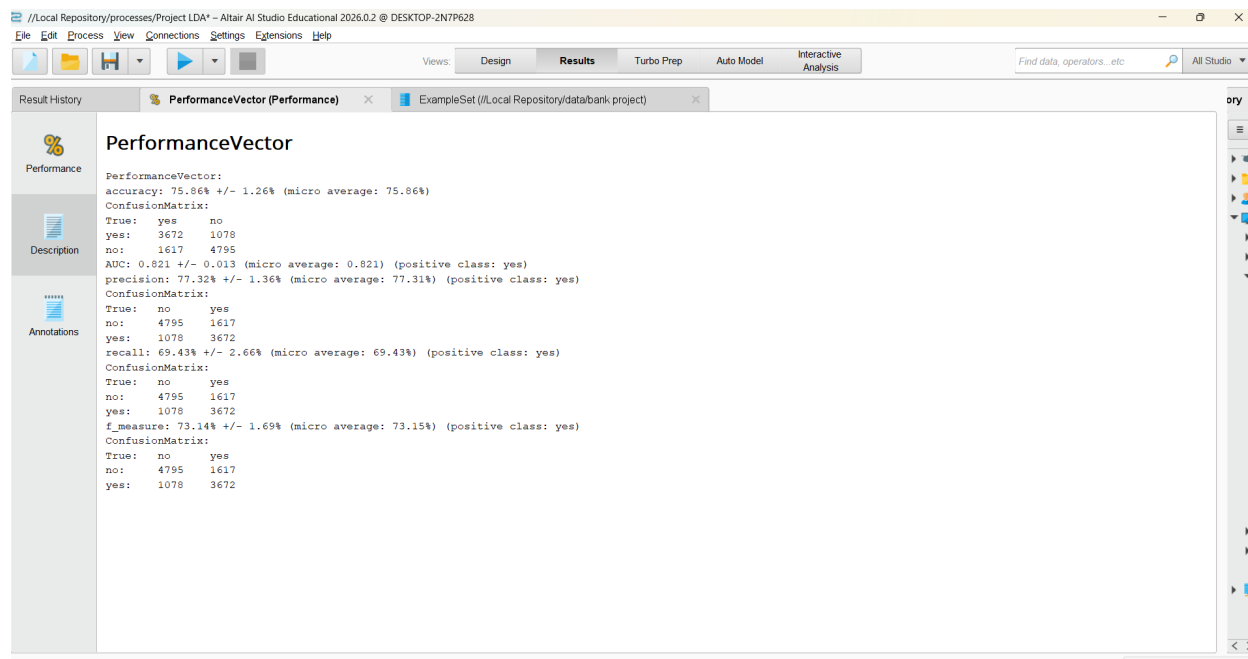
Cross Validation

- ☐ split on batch attribute
- ☐ leave one out
- number of folds: 10
- sampling type: stratified sampling
- ☐ use local random seed
- ☒ enable parallel execution

[Hide advanced parameters](#)

[Change compatibility \(12.0.002\)](#)

Help



Performance Result

The K - NN model achieved the following:

- Accuracy: 75.86%
- AUC: 0.821
- Precision: 77.32%
- Recall: 69.43%
- F1-score: 73.14%

Confusion Matrix

The model correctly classified a large number of customers, showing balanced performance.

Compared to LDA, K-NN improved both precision and recall, meaning it better identified potential subscribers while reducing errors.

K-NN performed better than simpler statistical models because it captures local patterns in the data. However, its performance depends heavily on distance calculations and data scaling.

Overall, K-NN showed good predictive performance, outperforming LDA and approaching Logistic Regression. It is effective for this dataset but may become inefficient with larger datasets due to computational cost.

Limitation of Model

- Sensitive to scaling of data
- Performance depends on k value
- Computationally expensive for large data
- Affected by noise and outliers

Naïve Bayes

Overview

Naïve Bayes was used as a probabilistic classification model to predict whether a customer will subscribe to a term deposit (deposit = yes/no). It is based on Bayes' Theorem and assumes that all features are independent of each other.

Methodology

The model was implemented using 10-fold cross-validation with stratified sampling to ensure consistent evaluation.

Data preprocessing steps included:

- Setting the label (deposit)
- Converting categorical variables (Nominal to Numerical)
- Normalizing numerical attributes

Inside cross-validation:

- Naïve Bayes model was trained
- Apply Model used for prediction
- Performance (Binominal Classification) used for evaluation

Hyper parameters

- Default Naïve Bayes settings
- Cross-validation: 10 folds
- Sampling: Stratified sampling
- No feature selection applied

Figure 7

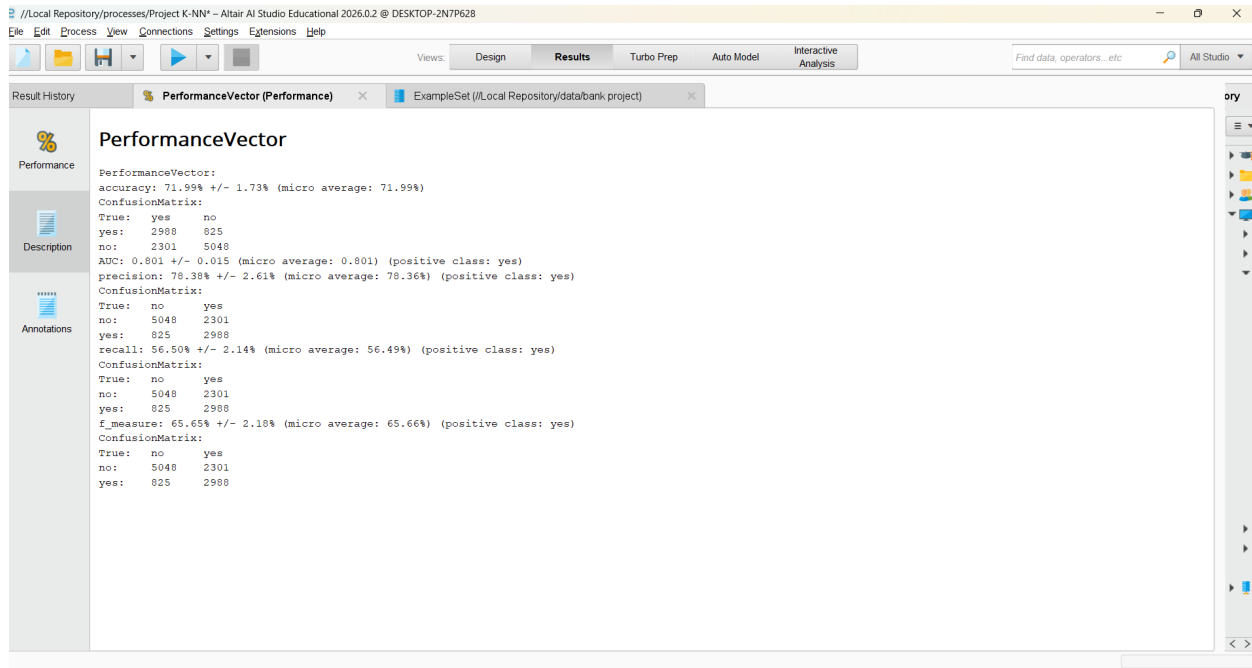
Rapid miner process for Naïve Bayes

The screenshot shows the Altair AI Studio interface with the following components:

- Repository:** A list of data sources including Airfares, Amtrak, bank project, Boston-Housing, Class Intro, customer-chum-data, ebay, and FlightDelays.
- Process:** A workflow diagram with the following steps:
 - Retrieve bank proj...** (Data source)
 - Set Role** (Operator)
 - Nominal to Numerical** (Operator)
 - Normalize** (Operator)
 - Cross Validation** (Operator)
- Parameters:** A panel on the right with settings for the **Process** step:
 - logverbosity: init
 - logfile: (empty)
 - resultfile: (empty)
 - random seed: 2001
 - send mail: never
 - encoding: SYSTEM
- Recommended Operators:** A list of operators with their usage percentages: Select Attributes (52%), Split Data (28%), Filter Examples (23%), and Multiply (23%).

The screenshot shows the Altair AI Studio interface with the following components:

- Repository:** A list of data sources including Airfares, Amtrak, bank project, Boston-Housing, Class Intro, customer-chum-data, ebay, and FlightDelays.
- Process:** A workflow diagram with the following steps:
 - Naive Bayes** (Operator)
 - Apply Model** (Operator)
 - Performance** (Operator)
- Parameters:** A panel on the right with settings for the **Cross Validation** step:
 - split on batch attribute: (unchecked)
 - leave one out: (unchecked)
 - number of folds: 10
 - sampling type: stratified sampling
 - use local random seed: (checked)
 - enable parallel execution: (checked)
- Recommended Operators:** A list of operators with their usage percentages: Select Attributes (52%), Split Data (28%), Filter Examples (23%), and Multiply (23%).



Performance Result

- Default Naïve Bayes settings
- Cross-validation: 10 folds
- Sampling: Stratified sampling
- No feature selection applied

Confusion Matrix

The model shows relatively high precision, meaning when it predicts a customer will subscribe, it is often correct. However, the recall is low, indicating that many actual subscribers are missing.

Naïve Bayes performs well due to its simplicity and efficiency, but its independence assumption limits its ability to capture relationships between variables. This results in lower recall compared to other models.

Overall, Naïve Bayes achieved moderate performance, with strong precision but weak recall. It is computationally efficient but less effective for complex datasets like bank marketing data.

Model Limitation

- Assumes independence between variables
- Cannot capture feature interactions
- Lower recall (misses subscribers)
- Simplistic model for complex data

Bagging

Overview

Random Forest is an ensemble classification model used to predict whether a customer will subscribe to a term deposit (deposit = yes/no). It builds multiple decision trees using bootstrap sampling (bagging) and random feature selection, and combines their predictions to improve accuracy and reduce overfitting.

Methodology

The Random Forest model was implemented in Rapid Miner using a structured machine-learning pipeline. The process began with retrieving the bank marketing dataset, followed by assigning the target variable (deposit) as the label.

Data preprocessing included converting categorical variables into numerical format using Nominal to Numerical and normalizing the data to ensure consistency across features.

The model was evaluated using 10-fold cross-validation with stratified sampling, which ensures that the distribution of the target variable is maintained across all folds and provides reliable performance estimates.

Within the cross-validation process:

- The Random Forest model was trained on training data
- The Apply Model operator was used to generate predictions
- Performance (Binominal Classification) was used to evaluate results

Hyper Parameters

Bootstrap sampling (bagging)

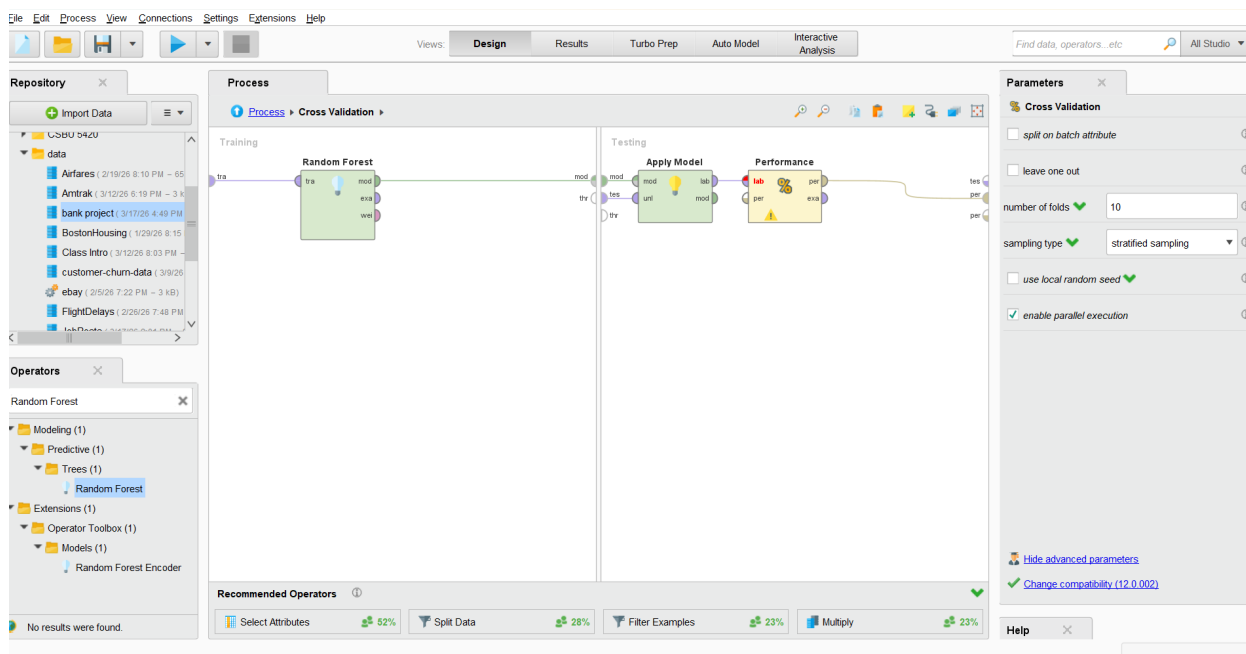
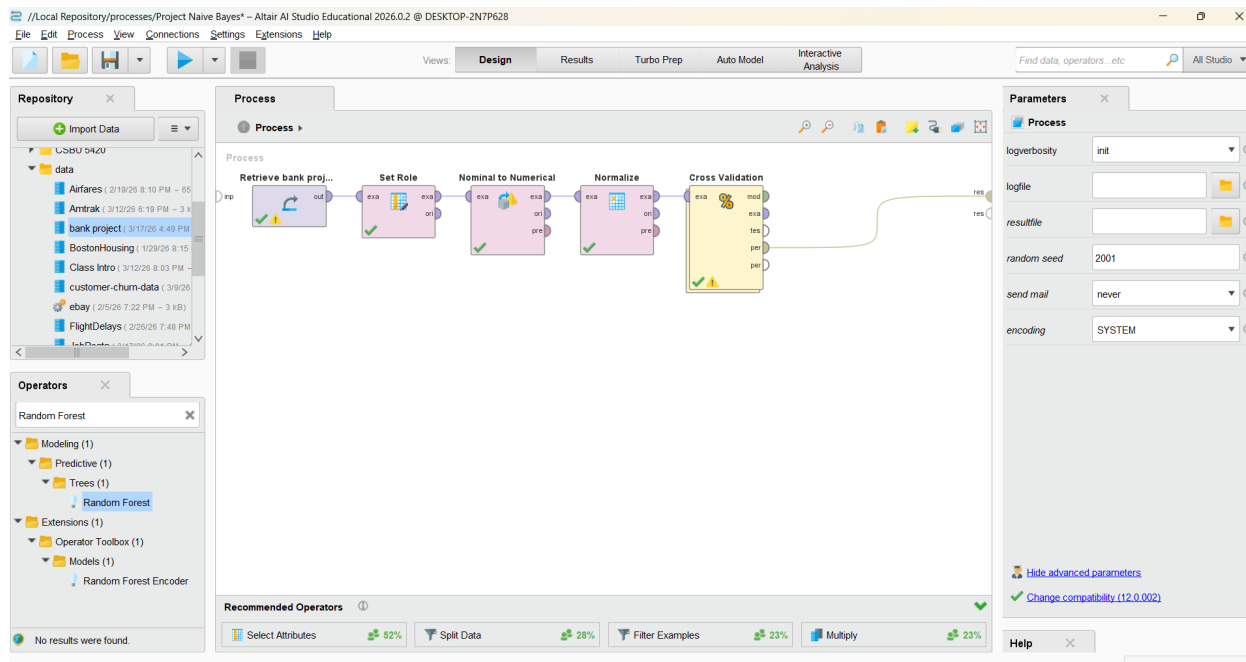
Random selection of features at each split

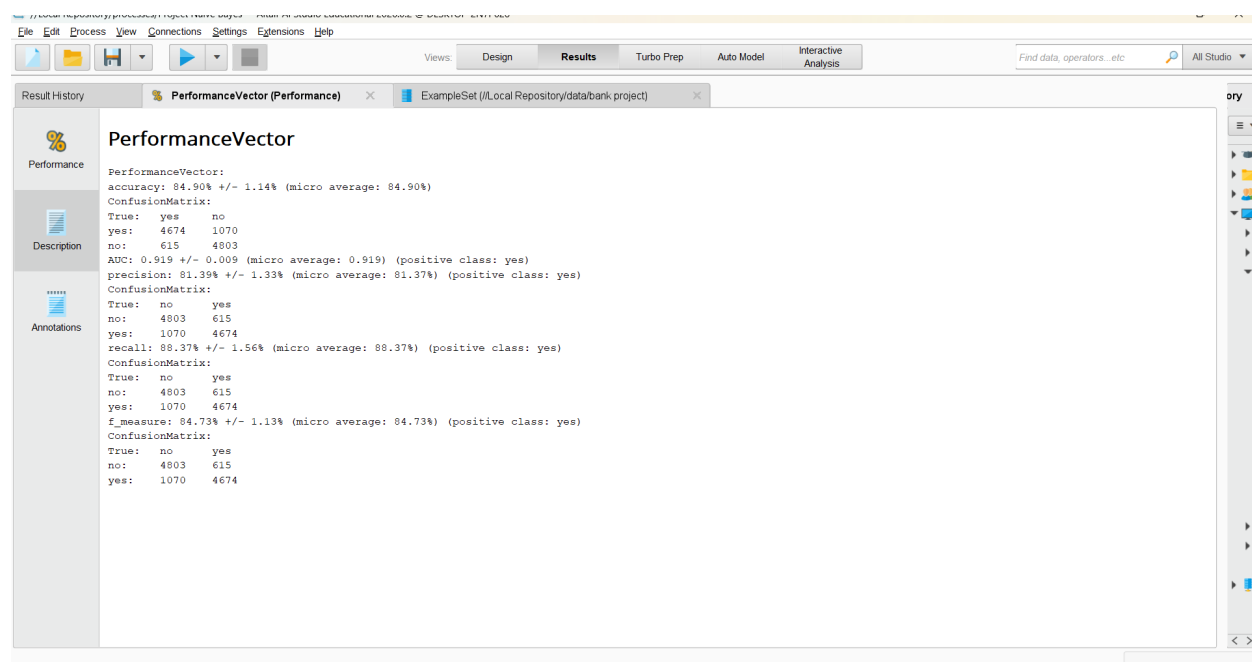
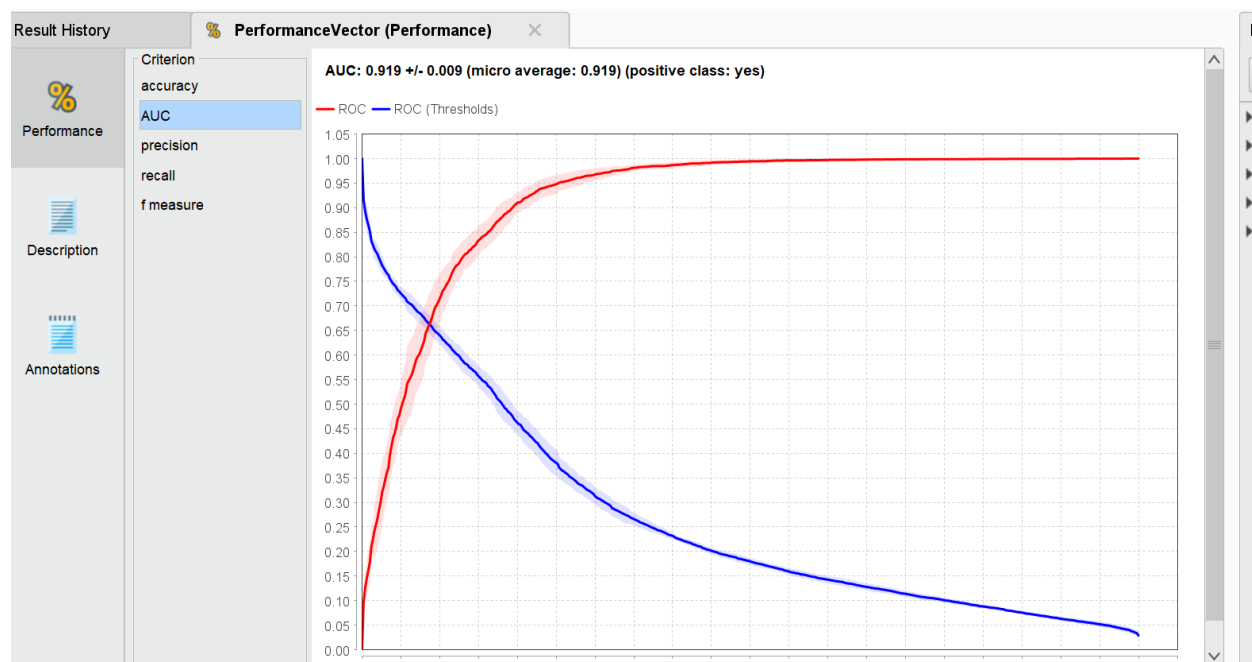
Cross-validation: 10 folds

Sampling type: Stratified sampling

Figure 8

Rapid miner process for Bagging





Performance Result

The Random Forest model achieved the highest performance among all tested models:

- Accuracy: 84.90%
- AUC: 0.919
- Precision: 81.39%
- Recall: 88.37%
- F1-score: 84.73%

Confusion Matrix

The model correctly classified a high number of both subscribers and non-subscribers. It shows strong recall, meaning it successfully identifies most customers who will subscribe, which is very important for marketing campaigns.

Random Forest performs significantly better because it:

- Captures complex relationships in data
- Reduces overfitting through averaging
- Handles both numerical and categorical features effectively

Overall, Random Forest (Bagging) achieved the highest performance among all models. It provides the best balance between accuracy, precision, and recall, making it the most suitable model for predicting customer subscription behavior.

Limitation of Model

- Less interpretable (black-box model)
- Computationally expensive
- Requires more resources

- May over fit if not tuned properly

Boosting

Overview

Boosting (AdaBoost) is an ensemble classification method used to predict whether a customer will subscribe to a term deposit (deposit = yes/no). It builds multiple weak learners sequentially, where each new model focuses on correcting the errors made by previous models to improve overall accuracy.

Methodology

The AdaBoost model was implemented in Rapid Miner using a structured pipeline. The process began with retrieving the bank marketing dataset, followed by assigning the target variable (deposit) as the label.

Data preprocessing steps included:

- Converting categorical variables using Nominal to Numerical
- Normalizing numerical attributes

The model was evaluated using 10-fold cross-validation with stratified sampling, ensuring consistent and reliable performance measurement.

Within cross-validation:

- A Decision Tree (max depth = 3) was used as the base learner
- AdaBoost combined multiple weak learners (iterations = 15)
- Apply Model was used for testing
- Performance (Binominal Classification) evaluated results

Hyper Parameters

- Algorithm: AdaBoost
- Base learner: Decision Tree
- Max depth: 3
- Iterations: 15
- Cross-validation: 10 folds
- Sampling: Stratified sampling

Figure 9

Rapid miner process for Boosting

Altair AI Studio Educational 2026.0.2 @ DESKTOP-2N7P628

File Edit Process View Connections Settings Extensions Help

Views: Design Results Turbo Prep Auto Model Interactive Analysis

Find data, operators, etc. All Studio

Repository

Import Data

data

- Airfares (2/19/26 8:10 PM - 85)
- Amtrak (3/12/26 6:19 PM - 3 k)
- bank project (3/17/26 4:49 PM)
- BostonHousing (1/29/26 8:15)
- Class Intro (3/12/26 8:03 PM)
- customer-churn-data (3/9/26)
- ebay (2/5/26 7:22 PM - 3 KB)
- FlightDelays (2/26/26 7:48 PM)

Operators

decis

Modeling (8)

- Predictive (8)
 - Trees (8)
 - Decision Tree
 - Random Forest
 - Gradient Boosted Trees
 - ID3
 - Decision Stump
 - Decision Tree (Multiclass)

We found "Weka Extension" in the Marketplace. [Show me!](#)

Process

Process

Retrieve bank proj... Set Role Nominal to Numerical Normalize Cross Validation

Drag here

Recommended Operators

- Select Attributes 52%
- Split Data 28%
- Filter Examples 23%
- Multiply 23%

Parameters

Process

logverbosity init

logfile

resultfile

random seed 2001

send mail never

encoding SYSTEM

[Hide advanced parameters](#)

[Change compatibility \(12.0.002\)](#)

Help

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Process

Process > Cross Validation

Training

AdaBoost

Testing

Apply Model Performance

Recommended Operators

- Select Attributes 52%
- Split Data 28%
- Filter Examples 23%
- Multiply 23%

Parameters

AdaBoost

iterations 15

Help

Double-click to enter subprocess, drag to move

The screenshot shows the Altair AI Studio interface in the Design view. The main workspace displays a workflow starting with a 'tra' data source, followed by a 'Decision Tree' model, and ending with a 'mod' output. The 'Parameters' panel on the right is configured for the 'Decision Tree' model with the following settings:

- criterion: gain ratio
- maximal depth: 3
- apply pruning: ☐
- apply prepruning: ☐

The 'Recommended Operators' panel at the bottom suggests several operators with their usage percentages:

- Select Attributes: 52%
- Split Data: 28%
- Filter Examples: 23%
- Multiply: 23%

The 'Repository' panel on the left shows a list of data sources, including 'Airfares', 'Amtrak', 'bank project', 'BostonHousing', 'Class Intro', 'customer-churn-data', 'ebay', and 'FlightDelays'. The 'Operators' panel shows a list of modeling operators, including 'Decision Tree', 'Random Forest', 'Gradient Boosted Trees', 'ID3', and 'Decision Stump'.

The screenshot shows the Altair AI Studio interface in the Results view. The main workspace displays the 'PerformanceVector' results for the 'Decision Tree' model. The performance metrics are as follows:

- accuracy: 76.98% +/- 2.32% (micro average: 76.98%)
- ConfusionMatrix:

True:	yes	no
yes:	3659	940
no:	1630	4933
- AUC: 0.819 +/- 0.018 (micro average: 0.819) (positive class: yes)
- precision: 80.56% +/- 5.44% (micro average: 79.56%) (positive class: yes)
- ConfusionMatrix:

True:	no	yes
no:	4933	1630
yes:	940	3659
- recall: 69.18% +/- 9.14% (micro average: 69.18%) (positive class: yes)
- ConfusionMatrix:

True:	no	yes
no:	4933	1630
yes:	940	3659
- f_measure: 73.84% +/- 3.42% (micro average: 74.01%) (positive class: yes)
- ConfusionMatrix:

True:	no	yes
no:	4933	1630
yes:	940	3659

Performance Result

The Boosting model achieved strong performance:

- Accuracy: 76.98%
- AUC: 0.819
- Precision: 80.56%
- Recall: 69.18%
- F1-score: 74.84%

Confusion Matrix

The model performs well in identifying subscribers with good precision. However, compared to Random Forest, the recall is lower, meaning some potential subscribers are still missing.

Boosting improves performance by focusing on difficult cases, but:

- It performs better than Naive Bayes and LDA
- Slightly better than K-NN in precision
- Still lower than Random Forest overall

AdaBoost provides strong predictive performance, but it does not outperform Random Forest.

While it improves accuracy over simpler models, it is not the best model for this dataset.

Limitation of Model

- Sensitive to noisy data and outliers
- Can over fit if too many iterations are used
- Sequential learning increases computation time
- Lower recall compared to Random Forest (misses some subscribers)

Table 1*Model Comparison Table*

Model	Accuracy	Precision	Recall	F1-Score	AUC	Performance Summary
Logistic Regression	82.57%	82.61%	84.73%	83.64%	0.903	Strong and Balanced Performance
LDA	71.27%	73.69%	65.51%	68.58%	1 (interpreted with caution)	Moderate, assumption based
K-NN	75.86%	77.32%	69.43%	73.14%	0.821	Good, Distance based
Naïve Bayes	71.99%	78.38%	56.50%	65.65%	0.801	High Precision, Low recall
Bagging	84.90%	81.39%	88.37%	84.73%	0.919	Best Overall performance
Boosting	76.98%	80.56%	69.18%	74.84%	0.819	Strong but below random forest

						(bagging)
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Random Forest (Bagging) achieved the highest Accuracy.

Performance Evaluation

Accuracy Metrics

Accuracy measures the proportion of correctly classified instances among all observations.

- Naive Bayes: 71.99%
- K-NN: 75.86%
- LDA: 71.27%
- Logistic Regression: 82.57%
- AdaBoost (Boosting): 76.98%
- Random Forest (Bagging): 84.90% (Highest)

Random Forest achieved the highest accuracy, indicating superior predictive performance.

Logistic Regression also showed strong performance, while Naïve Bayes and LDA performed lower due to their model assumptions.

Confusion Matrices

Confusion matrices summarize classification results by comparing actual and predicted values:

- True Positive (TP): Correctly predicted “yes”
- True Negative (TN): Correctly predicted “no”
- False Positive (FP): Incorrectly predicted “yes”
- False Negative (FN): Missed actual “yes”

Observations:

- Naive Bayes: High false negatives: misses many subscribers
- K-NN & LDA: Moderate performance with balanced errors
- Logistic Regression: Good balance of TP and TN
- AdaBoost: Improved predictions but still misses some positives
- Random Forest: Highest TP and TN with fewer FP and FN

Random Forest provides the most accurate classification with the lowest misclassification errors.

Its high recall makes it especially effective for identifying potential subscribers.

ROC and AUC Analysis

The ROC curve evaluates the model’s ability to distinguish between classes, while AUC summarizes overall performance.

- Naive Bayes: 0.801 (Good)

- K-NN: 0.821 (Good)
- LDA: 1.000 (Not reliable due to poor accuracy)
- Logistic Regression: 0.903 (Very Good)
- AdaBoost: 0.819 (Good)
- Random Forest (Bagging): 0.919 (Excellent)

Random Forest achieved the highest AUC, indicating excellent ability to distinguish between customers who will subscribe and those who will not. Logistic Regression also performed strongly, while Naïve Bayes and K-NN showed moderate discrimination ability.

Final Model and Insights

The Random Forest (Bagging) model emerged as the final model, which predicts customer subscription to term deposit accounts after all models underwent evaluation. The system achieved its highest accuracy result of 84.90% because of its strong AUC score of 0.919 and its ability to correctly identify 88.37% of cases. The results demonstrate that Random Forest effectively detects potential subscribers while showing equal proficiency in all measurement criteria.

Random Forest outperformed other models, which included Logistic Regression, K-NN, Naïve Bayes and LDA through its superior ability to analyze intricate data patterns. The ensemble method decreases overfitting while it boosts the system's ability to predict unfamiliar data. The analysis discovered that customer attributes, which include balance, job category and financial status, significantly determine their subscription choices.

The dataset contains distribution patterns that do not follow standard distribution and contains extreme value points, which ensemble models such as Random Forest, can more effectively manage. The model allows banks to reach their most promising customers, which results in better business outcomes. The process leads to better marketing results with decreased expenses and increased sales conversions. The project depends on Random Forest because it delivers the most dependable method for making data-driven marketing choices.

Conclusion

The project achieved its goals through the development and testing of various machine-learning models, which predicted customer subscription behavior for term deposits using the Bank Marketing Dataset. The research identified important customer subscription patterns through three processes which included data preprocessing, exploratory data analysis and model evaluation.

The Bagging Random Forest model achieved superior results compared to all other tested models because it produced the highest accuracy and AUC results while maintaining equal precision and recall measurements. The other models, which included Logistic Regression and KNN and LDA and Naïve Bayes and Boosting, provided valuable information but they could not match the predictive power of Random Forest.

The study results show that subscription behavior depends on three factors, which include customer balance, job category and previous customer interactions. Financial institutions use predictive customer identification tools to determine which customers will likely become interested in their products. This practice enables banks to optimize their marketing strategies, which results in better cost management and higher campaign success rates.

The project findings create important real world effects. Banks use machine-learning models for their marketing needs to move from existing outdated mass marketing methods toward modern data focused methods. This approach results in better customer interaction and increased rate of sales and more effective use of business assets.

Financial institutions should use Random Forest ensemble models according to the results of the study for their customer prediction needs. Future work should focus on improving performance through feature development and parameter optimization and advanced model implementation.

In conclusion, this project shows that machine learning is useful for solving real-world business issues and creates a firm background for implementing predictive analytics to influence judgement making in marketing.

Appendices

Appendix A: Detailed Calculation

The performance of each model was evaluated using standard classification metrics derived from the confusion matrix.

Formula

$$1. \text{Accuracy} = (TP + TN) / (TP + TN + FP + FN)$$

$$2. \text{Precision} = TP / (TP + FP)$$

$$3. \text{Recall (Sensitivity)} = TP / (TP + FN)$$

$$4. \text{F1-Score} = [2 * (\text{Precision} * \text{Recall})] / [\text{Precision} + \text{Recall}]$$

Appendix B: Additional Tables

Table B1

Model Performance Summary

Model	Accuracy	AUC
Logistic Regression	82.57%	0.903
LDA	71.27%	1 (interpreted with caution)
K-NN	75.86%	0.821
Naïve Bayes	71.99%	0.801
Bagging	84.90%	0.919
Boosting	76.98%	0.819

Table B2

Confusion Matrix Summary (Bagging)

	Predicted Yes	Predicted No
Actual Yes	4674(TP)	1070(FN)
Actual No	615(FP)	4803(TN)

References

Shmueli, G., Bruce, P. C., Deokar, A. V., & Patel, N. R. (2023). *Machine learning for business analytics: Concepts, techniques, and applications in RapidMiner*. John Wiley & Sons.

Bachmann, J. (n.d.). Bank marketing dataset. *Kaggle*.

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